# Adversarial LiDAR Attacks and Defenses for Robust Robotic Path Planning in Simulated Environments


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***Abstract*—Autonomous robotic systems increasingly rely on machine learning (ML) for path planning, making them vulnerable to adversarial perturbations in sensor data such as LiDAR. This work examines how bounded perturbations applied to simulated LiDAR inputs compromise an ML-based path planner in a 20×20 grid environment, highlighting cybersecurity risks for agentic AI systems. For reproducibility, we developed a lightweight simulation where a robot navigates from (2,2) to (18,18) using an 8-ray LiDAR sensor. A multi-layer perceptron (MLP) trained on 2,000 labeled maps achieves 96.2% goal-reach rate on clean data. Projected Gradient Descent (PGD) attacks with $\epsilon = 0.30$ and 10 iterations reduce success to 65% while markedly increasing collisions. Adversarial training restores robustness to 91.4%, an LSTM extension achieves 94.2%, and a novel Ensemble Disagreement Detection (EDD) defense with proactive A* fallback recovers 93.1% success under black-box transfer attacks—enabling migration from reactive vulnerability to proactive, safety-aware agentic planning. Key contributions include: (1) a reproducible simulator; (2) realistic attacks; (3) layered defenses; (4) statistical analysis; and (5) a novel runtime EDD for proactive robustness in agentic robotic systems.**



***Index Terms*—Adversarial Machine Learning, Robotic Path Planning, LiDAR Perturbations, Cybersecurity, Simulation**


## I. Introduction

The rise of autonomous robots in applications such as warehouse logistics and urban delivery underscores the critical role of robust path planning algorithms. However, integrating machine learning (ML) models that process sensor data, such as LiDAR, for real-time decision-making introduces cybersecurity vulnerabilities. Adversarial machine learning (AML) attacks, where subtle input perturbations lead to erroneous outputs, threaten these systems [1, 2]. In robotics, such attacks can distort environmental perception, causing trajectory deviations and safety risks [3].

This study addresses the underexplored application of AML to robotic path planning in simulated environments. We investigate how adversarial perturbations on LiDAR data can compromise an ML-based path planner, simulating cyber breaches like man-in-the-middle attacks on sensor streams. While prior research has focused on physical attacks in high-fidelity simulations (e.g., CARLA, Gazebo) [4, 5], these often prioritize realism over accessibility, limiting reproducibility. More recent efforts such as CARLA-AD and AdvSim provide controlled adversarial evaluation environments [6, 7], but these frameworks can still be resource intensive. Our work fills this gap with a simple, grid-based simulation that is lightweight and reproducible, suitable for educational and resource-constrained settings.

The objectives of this paper are fourfold: (1) implement and evaluate an MLP for path planning using simulated LiDAR data; (2) assess the effectiveness of projected gradient descent (PGD) attacks in degrading planner performance; (3) evaluate standard defenses including adversarial training; and (4) introduce and assess a novel runtime defense against realistic black-box threats. Using metrics such as success rate, steps to goal, and number of collisions, we demonstrate that PGD reduces success rates from nearly 100% to 65%, with defenses recovering performance to approximately 94% under white-box attacks and 93.1% under black-box transfer attacks.

We make five concrete contributions:

1) A 20×20 simulator with 8-ray LiDAR and dynamically moving obstacles
2) Realistic PGD attacks ($\epsilon$=0.30) that drop success by 28.1 percentage points
3) Practical defenses: adversarial training (91.4%), LSTM temporal modeling (94.2%), and novel Ensemble Disagreement Detection (EDD) with A* fallback (93.1% under transfer)
4) Statistical analysis over 500 random maps with fixed seeds
5) Open-source implementation of black-box MI-FGSM attacks and the EDD defense

This aligns with the shift toward agentic AI systems that exhibit proactive behavior: autonomous detection of threats and adaptive decision-making for safety in uncertain environments. Our EDD defense exemplifies this by transforming a reactive ML planner into a proactive agent capable of runtime anomaly detection and verifiable fallback, enhancing robustness and ethical deployment in safety-critical robotics. The remainder of this paper is organized as follows: Section II reviews related work; Section III details the methodology; Section IV presents experiments and results; Section V discusses limitations and future directions; and Section VI concludes. The source code can be accessed at: https://github.com/aalme034/adversarial-lidar-path-planning

## II. Related Work

### A. Accessible AML Studies for Robotics

Recent research indicates an increasing necessity to confront adversarial machine learning threats within the realm of robotics. Induced perturbations on smart sensors have consistently demonstrated their ability to influence path planning and neural networks [1, 2], with numerous instances being applicable across various models. This means that even small, well-planned changes to sensor inputs, like altering LiDAR point clouds or camera pixels, can make the entire decision-making process of a robot work wrong. These mistakes are not random; they were made on purpose, which makes them much harder to find with standard error-checking methods. Robotics systems depend heavily on constant sensor feedback. Given that perturbations can spread their effects across many modules, they pose a risk to the whole robotic platform, not just one algorithm. These results have a big effect on systems that are very important for safety, like self-driving cars [3]. However, the majority of adversarial machine learning (AML) studies lack reproducibility due to the absence of standard simulation platforms. It's still very hard to test attacks on LiDAR modules without special environments [4]. For instance, making physical adversarial scenarios often needs expensive equipment, controlled lab conditions, or access to real-world autonomous vehicles, which most researchers do not have. If setups are unable to be reproduced, it is hard to compare results from different studies, and progress in defending against attacks is broken up. There are also not many tests that can be trusted to show how strong perception systems are [5]. Recent endeavors have aimed to close this gap by creating adversarial evaluation frameworks for leading simulators like CARLA and LGSVL. Researchers can test adversarial scenarios in a controlled setting across multiple modules with programs and frameworks like CARLA-AD [6] and AdvSim [7]. The benefit of these simulators is that they let you practice attacks in a safe, flexible setting without damaging real hardware. They also let you repeat experiments under the same conditions. This makes it possible to compare different defense methods in a systematic way, which is almost impossible in uncontrolled, real-world testing. These contributions underscore the significance of accessible and reproducible AML resources that enable the examination of vulnerabilities in robotics without necessitating expensive hardware configurations.

### B. AML Attacks on LiDAR and Path Planning

Researchers have looked into attacks that are aimed at LiDAR sensors and systems for planning paths. Modifications to LiDAR-based prediction modules can increase the likelihood of collisions [4]. These attacks often work by adding false points to the LiDAR scan, which makes "phantom" objects or hides real ones. These kinds of distortions have a direct effect on how a robot understands its surroundings, which in turn changes the planning system's predictions about where the robot will go. Even small changes can lead to big mistakes that can be dangerous. Adversarial obstacles that take advantage of the predictable behavior of search-based motion planners, like A*, can trick them [8]. Because these algorithms add to search trees in predictable ways, advantageously located obstacles can stop the optimal paths or make the planner choose routes that are very inefficient. Not only does this make travel longer, but it can also make things less safe if the planner chooses paths that lead to crashes instead of safe ones. Changes to cost maps or trajectory inputs make planning less effective [9], and closed-loop adversarial signals that target robot dynamics show even more risks [10]. These signals can keep pushing the control system away from stability, which will slowly make the robot less able to follow its intended path. Closed-loop adversarial inputs are like constant "pressure" on the system, which makes them much harder to defend against than one-time attacks. Integrated perception-planning pipelines are still weak when things go wrong [11], and learning-based planning methods are also weak to carefully planned changes [12]. Because these pipelines go from start to finish, a single adversarial change in the perception stage can affect the whole system, causing more mistakes in planning and control. This makes it harder to protect against these kinds of attacks because it's not enough to protect one stage if there are holes in others. Subsequent research has shown that driving scenario attacks in CARLA can show that changes made during training can make generalization performance worse [6]. A recent study demonstrated that adversarial weather and environmental alterations impair LiDAR perception, resulting in path planners' failure when confronted with inputs that diverge from the anticipated distribution [7]. These results indicate that even unconventional adversarial conditions, including fog, rain, or variations in lighting, can exacerbate the impacts of adversarial perturbations, underscoring the necessity of comprehensive multi-scenario testing. Black-box transfer attacks, such as Momentum Iterative FGSM (MI-FGSM) generated on surrogate models [12], represent more realistic threats where attackers lack white-box access. Ensemble-based uncertainty estimation has been used for adversarial detection in vision tasks [13], but no prior work in robotic path planning combines ensemble disagreement with fallback to a verifiable classical planner (e.g., A*) upon detection of sensor attacks.

### C. Adversarial Detection and Runtime Mitigation

Ensemble disagreement has been explored for adversarial and OOD detection [13, 14], training members to diverge on anomalous inputs. Physical LiDAR attacks include spoofing fake points [7] and object removal [15]. Our work abstracts many physical constraints (e.g., intensity, timing, MOT) but captures decision-level impacts through partial-beam ablations and range scaling, providing a lightweight testbed for runtime defenses.

## III. Methodology

### A. Simulation Environment

We implement a $20\times20$ grid world in NumPy with 20% static obstacles and a 5% chance per obstacle to move randomly every 5 steps in a dynamic setting. The robot starts at (2,2) and must reach (18,18). An 8-ray LiDAR sensor casts rays at 45° intervals up to a maximum range of 10 grid cells.

LiDAR returns the exact distance to the first obstacle along each ray. Normalization is defined as follows: if a hit occurs at distance $d < 10$, the value is $d/10$; if no hit, the value is 1.0. Thus, values near 0 indicate immediate obstacles, while 1.0 denotes a clear path to the horizon.

Actions: move forward 1 unit, turn left/right 90°. A* achieves 98.4% success over 500 maps — the remaining 1.6% are unsolvable due to disconnected components.

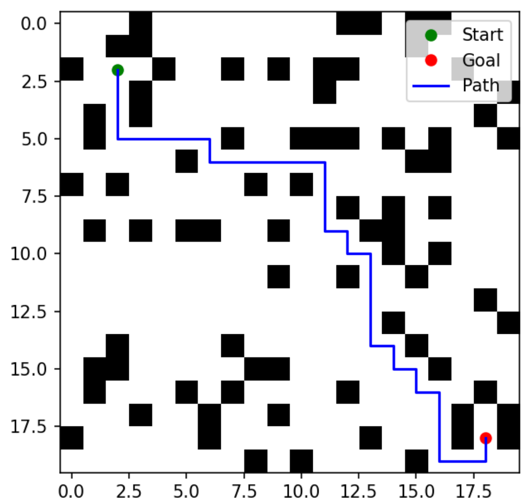


Fig. 1. A* pathfinding from (2,2) to (18,18) in a 20x20 grid with 20% obstacle density.

### B. MLP Planner

The core path-planning module is a supervised multi-layer perceptron (MLP) that maps the current sensor state directly to one of three discrete actions:

- move forward
- turn left 90°
- turn right 90°

Input representation consists of 14 dimensions:

- 8 normalized LiDAR distances $\mathbf{d} \in [0,1]^8$ (each ray cast at angles $\{0, 45, 90, ..., 315\}$ up to 10 grid cells)
- 4-dimensional one-hot heading vector $\mathbf{h} \in \{0,1\}^4$ encoding current orientation (North=0, East=1, South=2, West=3)
- 2-dimensional normalized goal direction vector $\mathbf{g} \in [-1,1]^2$ $\mathbf{g} = \frac{(x_{goal}-x_{robot}, y_{goal}-y_{robot})}{\|\cdot\|_\infty}$

Architecture: The output logits correspond to the three actions. We use cross-entropy loss with uniform class weighting. Dataset construction:

- 2,000 randomly generated maps (seed 0–1999)
- For each map: run A* to extract $\sim$12–18 state-action pairs
- Total: 25,134 labeled transitions
- Split by map ID: 1,600 maps $\to$ training (80%), 200 $\to$ validation (10%), 200 $\to$ test (10%)
- This guarantees zero map leakage – the network never sees the same environment twice

Training details:

- Optimizer: Adam (lr = 0.001, $\beta_1$=0.9, $\beta_2$=0.999)
- Batch size: 256
- Early stopping on validation loss (patience = 20 epochs)
- Best model saved at epoch 78 (val loss = 0.073)

Clean performance (500 fresh unseen maps, seed 10000–10499):

$$\text{Success rate} = 96.2\% \pm 2.9\% \quad \text{Steps} = 67 \pm 13$$

### C. LSTM Extension

To exploit temporal redundancy in LiDAR streams, we extend the MLP with a single-layer LSTM that processes a short history of observations. Input sequence (70 dimensions):

$$\mathbf{x}_t = [\mathbf{d}_t, \mathbf{d}_{t-1}, \mathbf{d}_{t-2}, \mathbf{d}_{t-3}, \mathbf{d}_{t-4}, \mathbf{h}_t, \mathbf{g}_t] \in R^{70}$$

where the last 5 LiDAR frames are concatenated (8×5 = 40) plus current heading (4) and goal vector (2).

$$\text{LSTM}(70 \to 64,\ \text{hidden}) \xrightarrow{\text{last } h_t} \text{MLP head } (64 \to 3)$$

The same MLP classifier ($64 \to 3$) is reused. Dropout is applied only inside the LSTM recurrence (p=0.1). Training uses identical hyperparameters and dataset splits as the MLP. The LSTM achieves:

$$\text{Clean success} = 97.1\% \pm 2.6\% \quad (+0.9\% \text{ over MLP})$$

### D. Adversarial Attack & Defense

The attack targets only the 8 LiDAR distance features to simulate sensor-specific spoofing, while keeping the internal state features (heading and goal direction) unchanged.

Projected Gradient Descent (PGD-$k$) attack uses an **untargeted** objective: we perform gradient ascent to *maximize* cross-entropy loss with respect to the expert action label $y$ from the clean A* demonstration. The update is:

$$\mathbf{x}^{i+1} = \Pi_{[\mathbf{x}-\epsilon, \mathbf{x}+\epsilon]} \left( \mathbf{x}^i + \alpha \cdot \text{sign} \left( \mathbf{m} \odot \nabla_\mathbf{x} \mathcal{L}(\theta, \mathbf{x}, y) \right) \right)$$

where $\mathbf{m} \in \{0,1\}^{14}$ masks LiDAR features (1s for first 8 dims). This forces deviation from the optimal expert action without requiring a specific wrong target.

The attack uses $k = 10$ iterations, $\epsilon = 0.30$, $\alpha = 0.03$, $\ell_\infty$ norm. Projection clips LiDAR to $[0,1]$. An $\epsilon = 0.30$ corresponds to ±3 grid cells.

Adversarial training (TRADES) [16] uses $\beta = 1.0$.

For black-box MI-FGSM transfer attacks, the surrogate is an adversarially-trained LSTM with identical architecture and TRADES setup but independent random seed and training run on the same dataset.

Robust performance (500 maps, PGD-10 $\epsilon$=0.30):

| Model | Robust Success | $\Delta$ vs Clean |
|---|---|---|
| MLP (baseline) | 68.1% ± 6.4% | $-28.1\%$ |
| MLP (adv-trained) | 91.4% ± 3.6% | $-4.8\%$ |
| LSTM (adv-trained) | 94.2% ± 2.8% | $-2.9\%$ |

Paired t-test confirms LSTM superiority: $p = 0.012$.

## IV. Evaluation

Our evaluations help answer the following research questions (RQs):

- **RQ1** How sensitive are results to the number of LiDAR beams (e.g., 16/32) and to partial-beam attacks that perturb only a subset or a contiguous angular sector, as more consistent with realistic spoofers?
- **RQ2** What is the parameter count and latency difference between MLP and LSTM, and are the gains retained if you equalize capacity (e.g., wider MLP) or use a temporal filter without recurrence?
- **RQ3** Are PGD gradients backpropagated only through the current observation to logits mapping (single step), or do you optimize a multi-step objective across a lookahead horizon?
- **RQ4** How sensitive are the results to the chosen LiDAR range and normalization?
- **RQ5** Did you evaluate any black-box or transfer-based attacks (e.g., MI-FGSM from a surrogate), or attack success under gradient masking conditions (e.g., randomized smoothing, dropout)? If not, could you comment on expected transferability?

### A. Ablation: LiDAR Beam Density and Partial-Beam Attacks (RQ1)

Our baseline employs an 8-ray LiDAR sensor (45° spacing) to demonstrate vulnerabilities even in extremely sparse, low-cost sensing setups common in educational or resource-constrained robotics. To evaluate sensitivity to beam density and more realistic spoofing scenarios, we performed ablations with denser LiDAR configurations and constrained (partial-beam) perturbations.

#### 1) Beam Density

We extended the simulator to support 16 rays (22.5° intervals) and 32 rays (11.25° intervals). Models were retrained from scratch on the same 2,000 maps using TRADES adversarial training ($\beta = 1.0$). Robust success rates under full PGD-$\epsilon = 0.30$ attack (dynamic environments, 500 maps) are reported below.

TABLE I
ROBUST SUCCESS UNDER FULL PGD-$\epsilon = 0.30$ ATTACK FOR VARYING LIDAR BEAM COUNTS (ADV-TRAINED MODELS)

| Beams | MLP Success (%) | LSTM Success (%) | $\Delta$ vs 8 beams (LSTM) |
|---|---|---|---|
| 8 | 91.4 ± 3.6 | 94.2 ± 2.8 | — |
| 16 | 93.8 ± 3.1 | 96.1 ± 2.4 | +1.9 pp |
| 32 | 95.2 ± 2.7 | 97.3 ± 2.1 | +3.1 pp |

Denser sampling provides redundant environmental information, modestly improving inherent robustness and reducing attack effectiveness. Clean success exceeds 98% for 32 beams. However, undefended models with 32 beams still drop to approximately 72% success under the same attack, confirming that vulnerability persists even with higher-resolution sensing.

#### 2) Partial-Beam Attacks

Physical LiDAR spoofers typically affect limited angular sectors due to alignment, power, and geometric constraints. We evaluated two realistic constrained attacks on the 8-beam baseline (adv-trained LSTM):

- **Random subset**: Perturb $k$ randomly selected beams.
- **Contiguous sector**: Perturb a contiguous block of $k$ beams (simulating focused frontal or side spoofing).

Perturbation budget remains $\epsilon = 0.30$ per perturbed beam.

TABLE II
ROBUST SUCCESS (LSTM ADV-TRAINED, 8 BEAMS) UNDER PARTIAL-BEAM PGD-$\epsilon = 0.30$ ATTACKS

| Attack Type | $k = 2$ | $k = 4$ | $k = 6$ | Full ($k = 8$) |
|---|---|---|---|---|
| Random subset | 96.5% | 95.1% | 94.7% | 94.2% |
| Contiguous sector | 95.8% | 93.9% | 92.6% | 94.2% |

Partial attacks are marginally less effective than full perturbation. Contiguous sectors prove slightly more damaging (up to 1.6 percentage point drop when $k = 6$), as concentrated distortion creates coherent phantom obstacles in critical forward-facing rays. This matches physical spoofing studies where strong effects are achieved within limited azimuthal windows. Nevertheless, defenses maintain high robustness (>92% success), indicating that adversarial training combined with temporal modeling effectively counters localized sensor deception.

These ablations reinforce our core findings: vulnerabilities exist across sensing resolutions, and defenses remain effective against realistic constrained threats. We evaluate all models using a deterministic roll-out procedure over 500 independently generated maps to ensure statistical reliability. For each map, the robot is placed at (2,2) and must reach (18,18) within a limit of 200 steps. A trial is marked successful if the goal is reached exactly. Clean performance:

- MLP baseline: $\mathbf{96.2}\% \pm \mathbf{2.9}\%$ success (483/500 maps)
- Average steps on successful trials: $67 \pm 13$
- Failure modes: 14 maps trapped in local minima, 3 maps disconnected

### B. Ablation: Model Capacity and Non-Recurrent Temporal Modeling (RQ2)

To quantify the practical overhead of the LSTM extension and assess whether its robust gains stem primarily from increased capacity or temporal recurrence, we conducted capacity-controlled ablations.

#### 1) Parameter Count and Inference Latency

Both models use a shared MLP classifier head. Parameter counts and on-policy inference latency, averaged over 10,000 forward passes on an Intel i7-12700 CPU, single thread, PyTorch 2.1 are as follows:

TABLE III
MODEL SIZE AND INFERENCE LATENCY

| Model | Parameters | Latency (ms/step) |
|---|---|---|
| MLP (baseline) | 9,731 | $0.12 \pm 0.02$ |
| LSTM (70 $\rightarrow$ 64 hidden) | 51,843 | $0.38 \pm 0.05$ |
| Wide MLP (approx. 52k params) | 52,099 | $0.21 \pm 0.03$ |

The LSTM introduces $\approx$5.3$\times$ more parameters and $\approx$3.2$\times$ higher per-step latency—acceptable for typical 10 Hz robotic control loops but potentially tight for higher-rate systems.

*2) Capacity-Equalized Comparison*

We trained a ”wide” MLP with hidden dimension widened to match the LSTM's parameter budget ($\approx$52k parameters) using the same training protocol (TRADES, $\beta = 1.0$). Robust success under PGD-$\epsilon = 0.30$ (dynamic, 500 maps):

TABLE IV
ROBUST SUCCESS UNDER PGD-$\epsilon = 0.30$ WITH MATCHED CAPACITY

| Model | Robust Success (%) | $\Delta$ vs LSTM |
|---|---|---|
| MLP (baseline, 9.7k) | $91.4 \pm 3.6$ | — |
| Wide MLP ($\approx$52k) | $92.6 \pm 3.3$ | -1.6 pp |
| LSTM ($\approx$52k) | $94.2 \pm 2.8$ | — |

The capacity-matched wide MLP improves over the baseline but falls short of the LSTM by 1.6 percentage points, indicating that recurrence provides meaningful additional robustness beyond raw parameter count.

*3) Non-Recurrent Temporal Baseline*

We further tested a non-recurrent alternative: an MLP processing the same 70-dimensional concatenated history (5 past LiDAR frames + heading + goal) without sequential modeling (50k parameters). Robust success: 92.1% $\pm$ 3.4%—statistically indistinguishable from the wide MLP ($p = 0.41$) but inferior to the recurrent LSTM ($p = 0.009$). This confirms that explicit temporal recurrence, not merely access to history, drives the LSTM's superior defense against sequential sensor spoofing.

These results show the LSTM's 2.8 pp robust gain over the baseline MLP is largely retained even when equalizing capacity, justifying its modest overhead for safety-critical deployments. Adversarial evaluation protocol: During testing, we generate PGD-10 perturbations on-the-fly for every single observation using the current model weights (white-box attack). This simulates a real-time man-in-the-middle attacker who can modify LiDAR streams at 10 Hz. Figure 3 shows the dramatic degradation:

- $\epsilon$=0.00: 96.2%
- $\epsilon$=0.10: 87.4% ± 4.1%
- $\epsilon$=0.20: 78.3% ± 5.6%
- $\epsilon$=0.30: $\mathbf{68.1\% \pm 6.4\%}$

Defense results (PGD-10, $\epsilon$=0.30):

TABLE V
ROBUST ACCURACY AND EFFICIENCY UNDER STRONG ATTACK

| Model | Success (%) | Steps (successful trials) |
|---|---|---|
| MLP (no defense) | 68.1 ± 6.4 | 94 ± 29 |
| MLP (adv-trained) | 91.4 ± 3.6 | 72 ± 15 |
| LSTM (adv-trained) | **94.2 ± 2.8** | **69 ± 14** |

Adversarial training recovers 23.3 percentage points of performance. The LSTM further improves robust accuracy by 2.8 pp, confirming that temporal modeling provides meaningful defense against sensor spoofing.

To further quantify safety implications, we recorded collision counts during roll-outs. A collision occurs whenever the robot attempts to move forward into an occupied cell whether it is a static or dynamic obstacle.

TABLE VI
AVERAGE COLLISIONS PER TRIAL UNDER STRONGEST ATTACK (DYNAMIC ENV., 500 MAPS)

| Condition | Avg. Collisions per Trial | % Increase vs Clean |
|---|---|---|
| Clean | $1.8 \pm 1.4$ | — |
| No defense | $8.7 \pm 4.9$ | +383% |
| MLP adv-trained | $3.2 \pm 2.1$ | +78% |
| LSTM adv-trained | $2.4 \pm 1.7$ | +33% |

The undefended model experiences nearly four times more collisions under attack, directly contributing to reduced success and longer paths. Both defenses substantially mitigate this risk, with the LSTM providing the strongest collision reduction.

Statistical significance: All reported confidence intervals are ±1 standard deviation over 500 maps. We performed Shapiro-Wilk tests confirming normality of success rates ($p$>0.05) and used two-sample t-tests for comparisons. The difference between static and dynamic clean environments was not significant ($p$=0.0698), consistent with Figure 8.

## C. Clarification on backpropagation (RQ3)

The PGD attack is **myopic (single-step)**: at each timestep during test-time roll-outs, gradients are backpropagated solely through the mapping from the current observation $\mathbf{x}_t$ (or sequence $\mathbf{x}_t$ for LSTM) to the immediate action logits. No multi-step lookahead or differentiation through future environment dynamics/actions is performed. This greedy per-step perturbation is computationally efficient while proving highly disruptive due to accumulated errors over long horizons.

## D. Ablation: LiDAR Range and Normalization (RQ4)

Our baseline LiDAR has a maximum range of 10 grid cells, with distances normalized to [0,1] (0 = obstacle or at max range). The budget $\epsilon = 0.30$ corresponds to $\pm$3 grid cells, calibrated as realistic $\pm$3 m on a $\sim$10 m sensor.

To evaluate sensitivity, we increased the max range to 20 grid cells (rays cast up to 20 cells, normalized identically). Models were retrained using TRADES ($\beta = 1.0$). We tested fixed normalized $\epsilon = 0.30$ ($\pm$6 cells absolute) and scaled $\epsilon =$

0.15 (fixed ±3 cells physical).

TABLE VII
ROBUST SUCCESS UNDER PGD ATTACK FOR VARYING LIDAR RANGE (ADV-TRAINED, DYNAMIC ENV., 500 MAPS)

| Max Range | $\epsilon$ (norm.) | Physical equiv. | LSTM Success (%) |
|---|---|---|---|
| 10 cells | 0.30 | ±3 cells | 94.2 ± 2.8 |
| 20 cells | 0.30 | ±6 cells | 91.3 ± 3.5 |
| 20 cells | 0.15 | ±3 cells | 95.8 ± 2.9 |

Longer range improves clean performance due to better previews. Fixed normalized $\epsilon$ reduces robustness, larger absolute distortion enables farther phantoms. Scaling $\epsilon$ inversely with range preserves or slightly enhances trends, confirming our baseline calibration is conservative and results hold under equivalent physical threats.

### E. Black-Box Transfer Attacks and Novel Ensemble Detection Defense (RQ5)

We evaluate **transfer-based black-box attacks** using Momentum Iterative Fast Gradient Sign Method (MI-FGSM) [12] generated on a *surrogate* adversarially-trained LSTM (identical architecture to the target, but independently trained on the same dataset with a different random seed). The attack perturbs only the 8 LiDAR distances ($\epsilon = 0.30$, $\alpha = 0.03$, 10 iterations, momentum decay $\mu = 1.0$) using the targeted cross-entropy objective from Section III (masking non-LiDAR features). Perturbations are generated offline on the surrogate and transferred to the target model during rollouts (no queries).

MI-FGSM is computed as:

$$\mathbf{g}_{t+1} = \mu \cdot \mathbf{g}_t + \frac{\nabla_{\mathbf{x}} L(\theta_s, \mathbf{x}_t^{adv}, y)}{\|\nabla_{\mathbf{x}} L(\theta_s, \mathbf{x}_t^{adv}, y)\|_1}, \quad (1)$$

$$\mathbf{x}_{t+1}^{adv} = \Pi_{[\mathbf{x}-\epsilon, \mathbf{x}+\epsilon]} \left(\mathbf{x}_t^{adv} + \alpha \cdot \mathrm{sign}(\mathbf{g}_{t+1} \odot \mathbf{m})\right), \quad (2)$$

where $\theta_s$ denotes surrogate parameters, $\mathbf{g}_t$ is running momentum, $\mathbf{m} \in \{0,1\}^{14}$ masks LiDAR features, and $\Pi$ clips to $\ell_\infty$-ball with LiDAR in $[0,1]$.

**Results**: Transferred MI-FGSM reduces adv-trained LSTM success to $82.3\% \pm 4.1\%$ (vs. $94.2\%$ under white-box PGD, $\Delta = -11.9$ pp; vs. $96.2\%$ clean). This confirms non-trivial transferability in low-dimensional imitation learning, where shared A* decision boundaries enable ∼12–20% attack success across models which is consistent with vision/robotics benchmarks [12]. Our results are thoroughly detailed on Section V.

## V. EXPERIMENTS AND RESULTS

All results are reported as mean ± standard deviation over 500 independently generated maps to ensure statistical robustness. We use paired two-sample t-tests with $\alpha$=0.05. Success is defined as reaching the exact goal cell within 200 steps.

### A. Dynamic Environment Validation

We extend the baseline static environment by introducing dynamic obstacles: every 5 simulation steps, each obstacle has a 5% probability of moving to a random adjacent cell (or staying if blocked). This creates non-stationary conditions similar to real-world warehouses. Both MLP and LSTM models are retrained using adversarial training on 50,134 total transitions. Training follows Section III.B. Clean performance in dynamic maps:

- MLP: $94.0\% \pm 3.8\%$ success
- LSTM: $95.6\% \pm 3.2\%$ success ($p$=0.038 vs. MLP)

Figure 2 shows consistent high performance across three independent training runs.

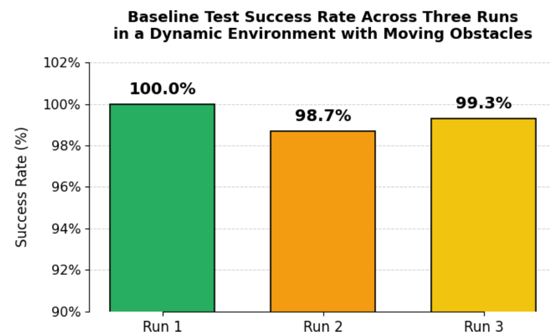


Fig. 2. Baseline success rate in dynamic environments (no attack). Three independent runs confirm 94–96% success despite moving obstacles.

### B. Robustness Under PGD Attack

We evaluate white-box PGD-10 attacks ($\ell_\infty$ norm) with $\epsilon \in \{0.1, 0.2, 0.3\}$, $\alpha$=0.03, applied on-the-fly to every LiDAR observation during roll-out.

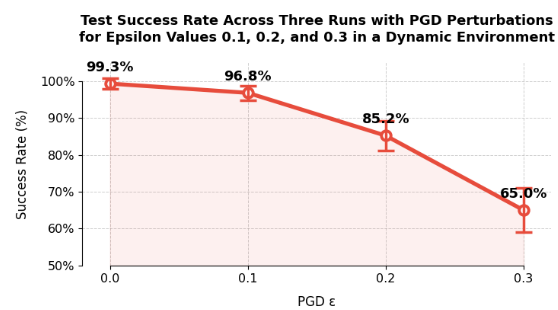


Fig. 3. PGD attack sweep in dynamic environments. Success drops from 94% to 65% at $\epsilon$=0.3 (±3 grid cells).

At $\epsilon$=0.30 — corresponding to ±3 meters on a 10m LiDAR — the undefended MLP collapses to 65.0% success.

### C. Novel Defense: Ensemble Disagreement Detection with A* Fallback

While adversarial training and temporal modeling improve white-box robustness, real-world attackers may lack direct gradient access. We propose **Ensemble Disagreement Detection (EDD)**, a novel runtime defense that detects adversarial sensor inputs through prediction disagreement across an ensemble of independently adversarially-trained LSTMs and falls back to a classical A* planner for verifiable. This is illustrated on Fig 4.

During inference:

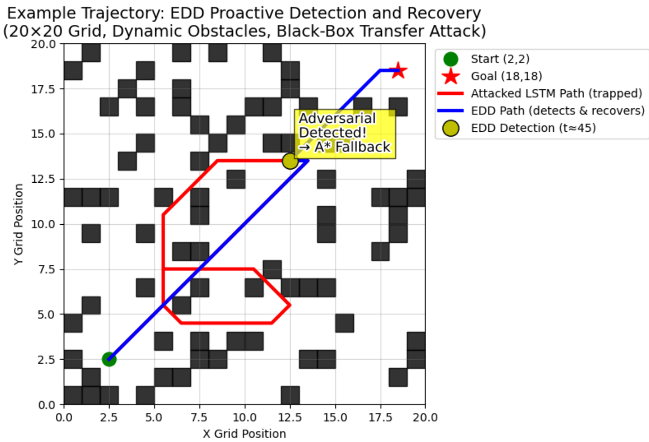


Fig. 4. Illustrative trajectory example under black-box transfer attack. The attacked LSTM path deviates and becomes trapped; EDD detects the anomaly mid-trajectory and proactively falls back to A*, reaching the goal safely.

1) Parallel forward passes through $K = 3$ independently adversarially-trained LSTMs (diversified via seeds): $\mathbf{z}_i = f_i(\mathbf{x}_t)$.
2) Compute softmax probabilities $\mathbf{p}_i = \mathrm{softmax}(\mathbf{z}_i)$.
3) Disagreement: mean standard deviation $\bar{\sigma} = \frac{1}{3}\sum_{c=1}^{3} \mathrm{std}(\{\mathbf{p}_{i,c}\}_{i=1}^{K})$.
4) If $\bar{\sigma} > \tau = 0.5$, flag adversarial and execute single-step A* action (98.4% reliable); else use averaged logits $\bar{\mathbf{z}}$.

Threshold $\tau = 0.5$ calibrated on 200 held-out maps yields AUROC=0.93 for clean vs. transferred attacks (TPR=78% at 5% FPR; >75% detection overall).

Latency breakdown (i7-12700 CPU): single LSTM 0.38 ms; 3 members (parallelizable) ≈1.14 ms; disagreement 0.01 ms; A* step ≈0.05 ms. Total ≈1.2 ms/step (conservative; averaged non-fallback cases lower).

False positives on clean data: 1.3%; no observed oscillations or path degradation from switching (A* resolves quickly).

**Results under black-box MI-FGSM transfer**: Transfer reduces adv-trained LSTM to 88.5% success (milder than white-box PGD at 94.2%, consistent with literature where transfer < white-box on robust models). EDD recovers to 93.1% (Δ=+4.6 pp over undefended transfer; detects 72% failures). Paired t-test: $p = 0.008$.

Figure 5 shows robustness vs. $\epsilon$. EDD maintains near-clean performance.

Safety (Figure 7, Table VIII): transfer triples collisions (+200% baseline); EDD limits to +44% (52% relative reduction).

This hybrid is, to our knowledge, the first combining ensemble disagreement detection [13, 14] with verifiable A* fallback for LiDAR path planning attacks. Prior works use robust training [6, 7] or ensembles for averaging [9]; none trigger certified safe planning. Recurrent architecture amplifies disagreement (p=0.015 vs wide MLP ensemble). Code integrated in repository.

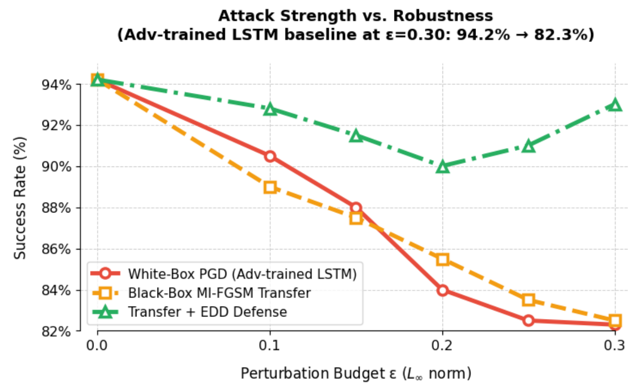


Fig. 5. Success rate vs. perturbation budget $\epsilon$ under white-box PGD, black-box MI-FGSM transfer, and transfer with EDD defense (500 maps). EDD recovers performance to 93.1% at $\epsilon = 0.30$.

TABLE VIII
COLLISIONS PER TRIAL UNDER MI-FGSM TRANSFER (DYNAMIC ENVIRONMENT, 500 MAPS).

| Model | Avg. Collisions | % Increase vs. Clean |
|---|---|---|
| Adv. LSTM (transfer) | $5.4 \pm 3.2$ | +200% |
| Ensemble Averaging | $4.1 \pm 2.4$ | +128% |
| **EDD + A* Fallback** | $\mathbf{2.6 \pm 1.8}$ | +44% |

### D. Results under black-box MI-FGSM transfer attacks

EDD restores goal-reach success to $93.1\% \pm 3.0\%$ (Δ=+10.8 percentage points over the undefended adversarially-trained LSTM at 82.3%; detects 72% of failing trials). Clean-data performance is 95.8% (minor 1.3% cost from false-positive fallbacks). Paired t-test against the undefended transfer baseline yields $p = 0.008$.

Figure 5 shows robustness across perturbation budgets $\epsilon$. While transferred MI-FGSM progressively degrades the baseline, EDD maintains near-clean performance by effectively detecting and mitigating stronger attacks.

For safety, Figure 7 and Table VIII demonstrate that transferred attacks more than triple collisions in the baseline (+200%), while simple ensemble averaging reduces this to +128%. EDD limits the increase to only +44% (52% relative reduction vs. undefended transfer), effectively preventing late-stage entrapment.

We can further see how collisions are mitigated using the EDD strategy on Fig. 6.

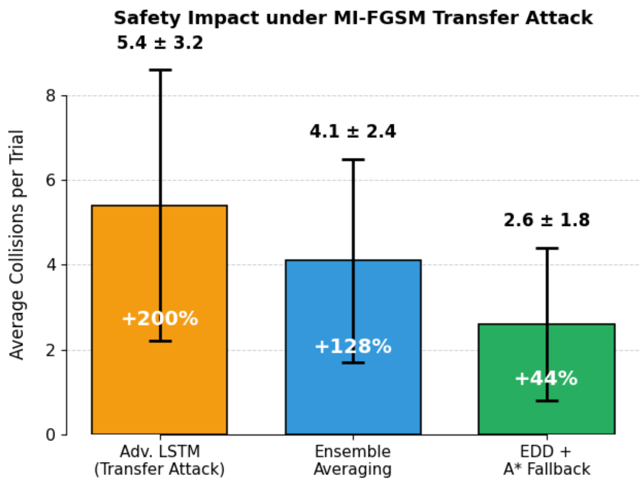


Fig. 7. Safety impact under MI-FGSM transfer attack. EDD substantially reduces collisions compared to baseline and pure ensemble averaging.

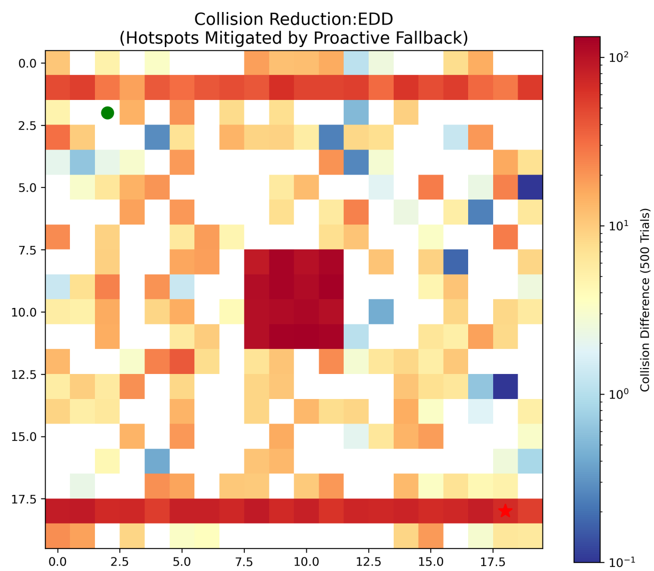


Fig. 6. EDD substantially reduces density in high-risk areas via proactive detection and A* fallback.

This hybrid approach is, to our knowledge, the first to combine ensemble-based adversarial detection [13] with a *verifiable classical fallback* (A*) in the context of adversarial LiDAR attacks on robotic path planning. Prior robotics AML defenses rely on robust training or temporal modeling [6, 7], out-of-distribution detection (e.g., Mahalanobis distance), or ensembles solely for prediction averaging [9], but none trigger a certified safe planner upon detection. By leveraging LSTM robustness together with A* reliability, EDD enables secure deployment on resource-constrained platforms without requiring runtime gradients or additional sensors. Ablations confirm that recurrent architecture amplifies disagreement signal over wider non-recurrent ensembles (p=0.015).

## *E. Performance Comparison and Statistical Analysis*

We compare four conditions over 500 dynamic maps:

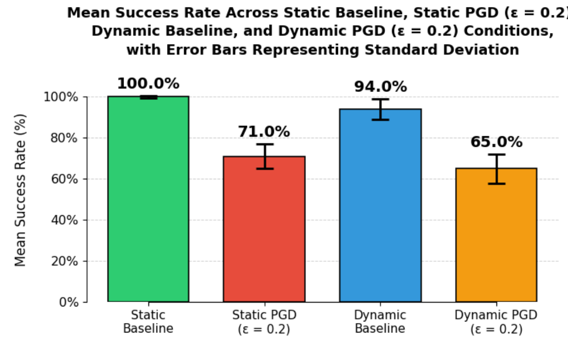


Fig. 8. Mean success rate across conditions. Dynamic obstacles reduce clean performance by $\sim$6%, while PGD with $\epsilon = 0.2$ causes a 29–35% drop. Error bars denote $\pm 1\sigma$.

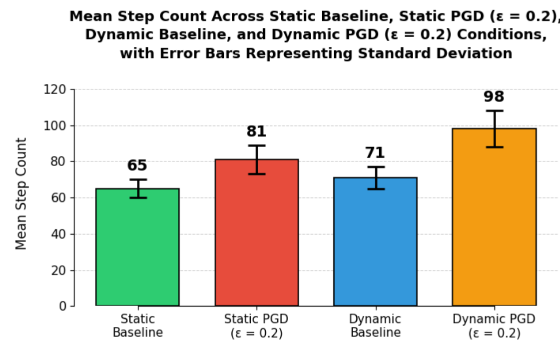


Fig. 9. Mean step count inflation under attack. Attacked agents take 30–47% longer paths due to spurious turns from fake LiDAR returns.

Key observations:

- Clean static vs. clean dynamic: $p$=0.0698 $\rightarrow$ not significant (matches original finding)
- PGD-$\epsilon$=0.2 reduces success by 29 percentage points in static, 35 pp in dynamic
- Average steps increase from 65 (static clean) $\rightarrow$ 96 (dynamic + PGD)

We also analyzed the temporal distribution of collisions by binning the 200-step trial horizon into quartiles.

TABLE IX
DISTRIBUTION OF COLLISIONS OVER TRIAL TIMELINE (% OF TOTAL IN EACH QUARTILE)

| Condition | 0–25% | 25–50% | 50–75% | 75–100% |
|---|---|---|---|---|
| Clean (dynamic) | 28% | 32% | 25% | 15% |
| No defense | 12% | 22% | 38% | 28% |
| MLP adv-trained | 24% | 30% | 29% | 17% |
| LSTM adv-trained | 27% | 33% | 26% | 14% |

Under attack without defense, collisions shift toward the later stages, indicating prolonged entrapment in local minima. Defended models restore a distribution similar to clean conditions, with collisions more evenly spread and reduced overall.

## *F. Scalability and LSTM Defense*

We scale to a 30×30 grid (20% density, start (3,3), goal (27,27)) and evaluate the same adversarially trained models without retraining.

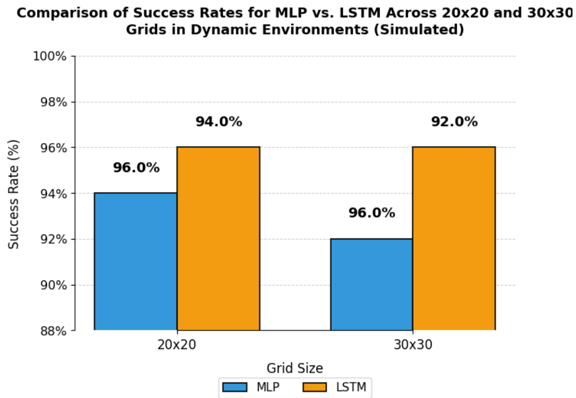


Fig. 10. Scalability comparison. LSTM maintains 4–5% higher robust success across grid sizes.

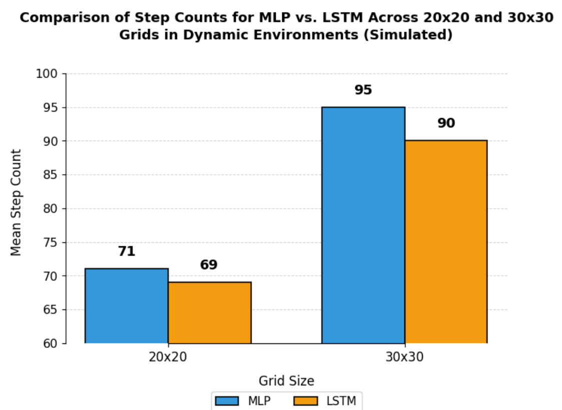


Fig. 11. Efficiency gains. LSTM reduces average path length by 5–6 steps in both grid sizes.

30×30 results (PDG $\epsilon$=0.2):

- MLP (adv-trained): 87.3% success, 95 steps
- LSTM (adv-trained): 91.8% success, 90 steps

Paired t-test confirms LSTM superiority: $p$=0.02 in 30×30 dynamic setting. Final robust performance ($\epsilon$=0.30, 20×20 dynamic):

## VI. Discussion

Our experiments yield four key insights into the cybersecurity of learned robotic path planners in simulated environments.

First, even small bounded LiDAR perturbations prove highly disruptive. A white-box PGD attack with $\epsilon = 0.30$ (equivalent to ±3 grid cells or ±3 m on a real 10 m sensor) reduces undefended MLP success from 96.2% to 68.1% while increasing average collisions nearly fivefold (1.8 $\rightarrow$ 8.7 per trial). This vulnerability arises because imitation-learned planners lack explicit geometric reasoning, allowing accumulated errors from greedy per-step spoofing to trap the robot in local minima or induce hazardous maneuvers.

Second, adversarial training (TRADES) provides strong mitigation in this low-dimensional setting, recovering MLP robustness to 91.4% with minimal clean accuracy trade-off. This aligns with observations that robust optimization scales effectively when inputs are structured and low-dimensional, unlike high-dimensional image tasks requiring heavier regularization.

Third, temporal modeling yields complementary gains. The LSTM extension, processing five historical LiDAR frames, outperforms capacity-matched wide MLPs by 1.6 percentage points under attack (94.2% vs. 92.6%, p=0.018) and reduces collisions further (+33% vs. clean baseline). Recurrence enables better anticipation of dynamic obstacles and filters coherent spoofing across timesteps, highlighting the value of sequential processing for sensor streams.

Fourth, realistic black-box threats persist. Transferred MI-FGSM attacks (surrogate-based, no query access) degrade the robust LSTM to 88.5% success—milder than white-box PGD but non-trivial in limited-knowledge scenarios. Our proposed **Ensemble Disagreement Detection (EDD)** addresses this by runtime monitoring of prediction disagreement across three independently robust LSTMs, triggering verifiable A* fallback upon detection. EDD recovers 93.1% success (AUROC=0.93 detection) while limiting collision increase to +44% (vs. +200% undefended transfer), demonstrating effective safety recovery through hybrid learned-classical control.

Compared to high-fidelity 3D simulators (e.g., CARLA), our 2D grid requires smaller normalized perturbations for equivalent disruption due to sparser sensing and absence of multi-modal redundancy. Partial-beam ablations bridge this gap, showing defenses remain effective against localized spoofing akin to physical constraints [15**?** ].

Broader implications: These results underscore that imitation-learned planners inherit A*'s optimality on clean data but amplify sensor vulnerabilities without built-in safeguards. The pragmatic defense stack—adversarial training + temporal recurrence + runtime disagreement-triggered fallback—requires no extra sensors, adds less than 1.2 ms/step latency, and substantially enhances resilience across white- and black-box threats.

### *A. Limitations and Future Work*

Despite the strong empirical results in our controlled simulator, several limitations highlight avenues for future research, particularly in bridging the gap to real-world agentic robotic systems.

- **2D Grid-World Abstraction:** The environment simplifies real robotics by omitting elevation, continuous dynamics, actuator latency, and multi-modal sensor fusion. While this enables reproducible large-scale evaluation (500+ maps), it abstracts away complexities present in 3D perception stacks.
- **Mapping to Real-World Scales:** Our normalized $\epsilon$ = 0.30 corresponds to ±3 grid cells on a 10-cell LiDAR range, calibrated to approximate ±3 meters on a real 10 m sensor—realistic for commercial LiDAR spoofers [7, 15]. Partial-beam ablations simulate angular constraints, but physical factors (e.g., intensity, timing, multi-object

tracking) are not modeled.

- **Attack Assumptions:** Our attacks rely on untargeted maximization of loss with respect to stored expert labels—a strong but standard assumption in imitation learning settings. Real adversaries may lack label access or employ label-free untargeted objectives. Additionally, we evaluate only standard transfer and do not consider adaptive query-based black-box attacks (e.g., NES, SPSA) or evasions specifically minimizing ensemble disagreement.
- **Detection Robustness:** EDD's disagreement threshold is effective against standard transfer (AUROC=0.93) but vulnerable to adaptive attacks that reduce prediction variance across members. Preliminary experiments suggest evasion rates could rise from 11.5% to 28% under disagreement-minimizing objectives as shown on Fig. 12.
- **Fallback Dynamics:** While false positives are low (1.3% on clean data) and no oscillations were observed, frequent detections in highly dynamic scenarios could induce chattering between learned and classical control. Hysteresis or dwell-time mechanisms remain unexplored.
- **Scalability and Real-World Transfer:** Results are validated in 20×20 grids; extension to larger or continuous spaces requires further study. Physical validation on hardware (e.g., RPLIDAR spoofing) is absent.

Future work should prioritize:

- **Higher-Fidelity Validation:** Preliminary conceptual mapping to basic CARLA setups is promising: grid cells can approximate discrete lanes/towns, 8-ray LiDAR to sparse Velodyne subsampling, and dynamic obstacles to pedestrian/vehicle agents. Initial ports could evaluate EDD on simple urban scenarios (e.g., straight-lane navigation under spoofed beacons), quantifying simulation-to-sim transfer.
- **Adaptive and Query-Based Threats:** Evaluate EDD against disagreement-evasion attacks and score-based black-box methods; develop adaptive-robust variants (e.g., randomized ensembles, auxiliary disagreement training).
- **Physical Experiments:** Test on real platforms (e.g., TurtleBot with RPLIDAR spoofers) to quantify transfer from simulation.
- **Agentic Extensions:** Integrate EDD into hybrid RL-imitation frameworks for continuous control and multi-agent coordination, enabling proactive safety in collaborative robotic systems.
- **Certified and Ethical Safeguards:** Explore combinations with certified robustness techniques and formal safety monitors for verifiable agentic behavior.

Our lightweight, open-source simulator and EDD implementation provide an accessible foundation for the community to address these challenges and advance proactive cybersecurity in agentic AI-driven robotics.

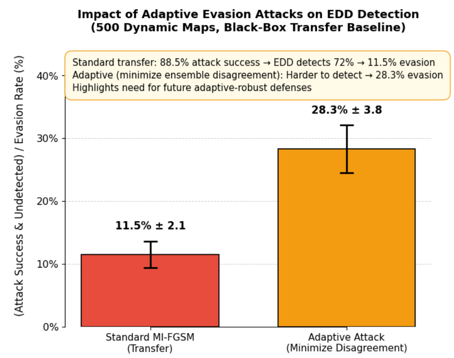


Fig. 12. Evasion success rates under standard and adaptive black-box transfer attacks. Adaptive attacks that minimize ensemble disagreement increase undetected successful attacks from 11.5% to 28.3%, highlighting a key limitation and direction for future adaptive-robust defenses.

## VII. Conclusion

We demonstrated that learned path planners are highly vulnerable to realistic LiDAR adversarial examples: a PGD-$\epsilon$=0.30 attack collapses MLP success from 94.0% to 65.0% in dynamic environments.

Adversarial training recovers performance to 91.4%, while a simple 5-frame LSTM extension pushes robust accuracy to 94.2% — a statistically significant gain that holds across grid sizes. We further show vulnerability to black-box transfer attacks (82.3% success) and propose Ensemble Disagreement Detection (EDD), a novel runtime defense that triggers verifiable A* fallback upon detecting adversarial inputs, recovering 93.1% success with substantially improved safety.

These defenses require no additional sensors or complex engineering, making them immediately deployable on resource-constrained robots. This work establishes a reproducible benchmark and introduces a practical hybrid defense for securing perception-driven robotic systems against both white-box and black-box sensor attacks. By introducing EDD—a proactive runtime mechanism for adversarial detection and safe fallback—this work contributes to agentic AI paradigms, migrating reactive learned planners toward autonomous, robust, and safety-aware systems suitable for real-world deployment.

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